

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414273
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Bean, Jr.; John H.

Immersion cooling reservoir level control

Abstract

Various embodiments include first and second coolant level regulator components of first and second component cooling tanks and a processor. Each of the first and second coolant level regulator components may include a temperature sensor configured to measure a temperature of the coolant in a reservoir, a coolant release valve configured to regulate the release of coolant from the reservoir, and a coolant level detector configured to detect a level of the coolant in the reservoir. The processor may receive temperature measurements from the temperature sensors, signal a target coolant level update for adjusting a level of the coolant in the reservoir by the coolant release valve. The target coolant level update may be determined based on a temperature average of the received temperature measurements from the temperature sensors. The coolant release valve may adjust a release of the coolant from the reservoir based on the target coolant level update.

Inventors:	Bean, Jr.; John H. (Littlestown, PA)
Applicant:	Green Revolution Cooling, Inc. (Austin, TX)
Family ID:	91953250
Assignee:	Green Revolution Cooling, Inc. (Austin, TX)
Appl. No.:	18/101478
Filed:	January 25, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240251531 A1	Jul. 25, 2024

Publication Classification

Int. Cl.: H05K7/20 (20060101)

U.S. Cl.:

CPC **H05K7/20836** (20130101); **H05K7/20236** (20130101); **H05K7/20763** (20130101);

Field of Classification Search

CPC: H05K (7/20236); H05K (7/20763); H05K (7/20836)

USPC: 700/300

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1093951	12/1913	Steinberger	N/A	N/A
1606025	12/1925	Gruendler	N/A	N/A
2440930	12/1947	Guglielmo et al.	N/A	N/A
2525457	12/1949	Paluev	N/A	N/A
2643282	12/1952	Greene	N/A	N/A
3094133	12/1962	Treanor	N/A	N/A
3320762	12/1966	Murdoch	N/A	N/A
3406244	12/1967	Sevgin	N/A	N/A
3450265	12/1968	Kreusch et al.	N/A	N/A
3489207	12/1969	Miller	N/A	N/A
3600636	12/1970	Petersen	N/A	N/A
3754741	12/1972	Whitehurst et al.	N/A	N/A
3858090	12/1973	Lehmann	N/A	N/A
4034335	12/1976	Harazoe et al.	N/A	N/A
4245668	12/1980	Lindstrom	N/A	N/A
4302793	12/1980	Rohner	N/A	N/A
4313310	12/1981	Kobayashi et al.	N/A	N/A
4372632	12/1982	Villa et al.	N/A	N/A
4399501	12/1982	Masselin	N/A	N/A
4460008	12/1983	O'Leary et al.	N/A	N/A
4464315	12/1983	O'Leary	N/A	N/A
4493010	12/1984	Morrison et al.	N/A	N/A
4590538	12/1985	Cray, Jr.	N/A	N/A
4600230	12/1985	Ise	N/A	N/A
4618268	12/1985	Horner	374/102	G01K 13/10
4648043	12/1986	O'Leary	N/A	N/A
4659459	12/1986	O'Leary et al.	N/A	N/A
4704658	12/1986	Yokouchi et al.	N/A	N/A
4722577	12/1987	Mewborne	N/A	N/A
4741385	12/1987	Bergles et al.	N/A	N/A
4834257	12/1988	Book et al.	N/A	N/A
5102503	12/1991	Silinski et al.	N/A	N/A
5145585	12/1991	Coke	N/A	N/A
5260850	12/1992	Sherwood et al.	N/A	N/A

5294916	12/1993	Bolton et al.	N/A	N/A
5297621	12/1993	Taraci et al.	N/A	N/A
5329418	12/1993	Tanabe	N/A	N/A
5332494	12/1993	Eden et al.	N/A	N/A
5414591	12/1994	Kimura et al.	N/A	N/A
5574627	12/1995	Porter	N/A	N/A
5655459	12/1996	O'Connor et al.	N/A	N/A
5851143	12/1997	Hamid	N/A	N/A
5907473	12/1998	Przilas et al.	N/A	N/A
6019167	12/1999	Bishop et al.	N/A	N/A
6374627	12/2001	Schumacher et al.	N/A	N/A
D477617	12/2002	Resch	N/A	N/A
6600656	12/2002	Mori et al.	N/A	N/A
6616851	12/2002	Sforza-Heinen et al.	N/A	N/A
6621707	12/2002	Ishimine et al.	N/A	N/A
D483045	12/2002	Resch	N/A	N/A
6909606	12/2004	Barsun et al.	N/A	N/A
7086247	12/2005	Campbell et al.	N/A	N/A
D530346	12/2005	Resch	N/A	N/A
7184269	12/2006	Campbell et al.	N/A	N/A
7210304	12/2006	Nagashima et al.	N/A	N/A
7278273	12/2006	Whitted et al.	N/A	N/A
7307841	12/2006	Berlin et al.	N/A	N/A
7318322	12/2007	Ota et al.	N/A	N/A
7403392	12/2007	Attlesey et al.	N/A	N/A
7413394	12/2007	Risser	N/A	N/A
7511959	12/2008	Belady et al.	N/A	N/A
7511960	12/2008	Hillis et al.	N/A	N/A
7551971	12/2008	Hillis	N/A	N/A
7604027	12/2008	Thatcher	N/A	N/A
7609518	12/2008	Hopton et al.	N/A	N/A
7686175	12/2009	Waisanen	N/A	N/A
7724513	12/2009	Coglitore et al.	N/A	N/A
7724517	12/2009	Attlesey et al.	N/A	N/A
7757506	12/2009	Ellsworth, Jr. et al.	N/A	N/A
7843298	12/2009	Hosokawa et al.	N/A	N/A
7854652	12/2009	Yates et al.	N/A	N/A
7856838	12/2009	Hillis et al.	N/A	N/A
7905106	12/2010	Attlesey	N/A	N/A
7911782	12/2010	Attlesey et al.	N/A	N/A
7911793	12/2010	Attlesey	N/A	N/A
7934386	12/2010	Rummel et al.	N/A	N/A
7961463	12/2010	Belady et al.	N/A	N/A
7971632	12/2010	Eriksen	N/A	N/A
7983040	12/2010	Campbell et al.	N/A	N/A
8009419	12/2010	Attlesey et al.	N/A	N/A
8047904	12/2010	Yates et al.	N/A	N/A
8213261	12/2011	Imhof et al.	N/A	N/A
8295047	12/2011	Hamburgen et al.	N/A	N/A

8310829	12/2011	Monk et al.	N/A	N/A
8654529	12/2013	Attlesey	N/A	N/A
8817465	12/2013	Campbell et al.	N/A	N/A
8964391	12/2014	Campbell et al.	N/A	N/A
9042098	12/2014	Campbell et al.	N/A	N/A
9049800	12/2014	Shelnutt et al.	N/A	N/A
D744996	12/2014	Keisling et al.	N/A	N/A
9269544	12/2015	Koo et al.	N/A	N/A
9386727	12/2015	Barringer et al.	N/A	N/A
9439327	12/2015	Sheng et al.	N/A	N/A
9504190	12/2015	Best	N/A	N/A
9664180	12/2016	Saavedra	N/A	N/A
9699938	12/2016	Shelnutt et al.	N/A	N/A
D796654	12/2016	Baker et al.	N/A	N/A
9756766	12/2016	Best	N/A	N/A
9801465	12/2016	Finch, Jr.	N/A	N/A
9861194	12/2017	Park	N/A	N/A
D821326	12/2017	Kreiner	N/A	N/A
9992914	12/2017	Best et al.	N/A	N/A
10123463	12/2017	Best et al.	N/A	N/A
10212849	12/2018	Matsumoto et al.	N/A	N/A
10306804	12/2018	Chester et al.	N/A	N/A
D852936	12/2018	Baker et al.	N/A	N/A
10342164	12/2018	So et al.	N/A	N/A
10405457	12/2018	Boyd et al.	N/A	N/A
D875206	12/2019	Gourgel et al.	N/A	N/A
10667427	12/2019	Gao	N/A	N/A
10743438	12/2019	Wakino et al.	N/A	N/A
10820446	12/2019	Boyd et al.	N/A	N/A
10939581	12/2020	Chen et al.	N/A	N/A
11000943	12/2020	Reynolds	N/A	N/A
11359865	12/2021	Stone et al.	N/A	N/A
11369041	12/2021	Yang et al.	N/A	N/A
11457735	12/2021	D'Lsep	N/A	N/A
11516943	12/2021	Bilan et al.	N/A	N/A
2002/0014460	12/2001	McKay	N/A	N/A
2002/0080575	12/2001	Nam et al.	N/A	N/A
2002/0151799	12/2001	Pantages et al.	N/A	N/A
2002/0185262	12/2001	Baer	N/A	N/A
2002/0189173	12/2001	Staschik	N/A	N/A
2003/0053293	12/2002	Beitelmal et al.	N/A	N/A
2003/0127240	12/2002	Beckbissinger et al.	N/A	N/A
2004/0008490	12/2003	Cheon	N/A	N/A
2004/0013563	12/2003	Romer et al.	N/A	N/A
2004/0050491	12/2003	Miya et al.	N/A	N/A
2004/0223300	12/2003	Fink et al.	N/A	N/A
2004/0246683	12/2003	Honsberg-Riedl et al.	N/A	N/A
2004/0254682	12/2003	Kast	N/A	N/A

2005/0011839	12/2004	Dart et al.	N/A	N/A
2005/0024826	12/2004	Bash et al.	N/A	N/A
2005/0052847	12/2004	Hamman	N/A	N/A
2005/0083657	12/2004	Hamman	N/A	N/A
2005/0111184	12/2004	Cliff et al.	N/A	N/A
2005/0114876	12/2004	Atarashi et al.	N/A	N/A
2005/0152112	12/2004	Holmes et al.	N/A	N/A
2005/0259402	12/2004	Yasui et al.	N/A	N/A
2006/0026610	12/2005	Sasao et al.	N/A	N/A
2006/0064709	12/2005	Throckmorton et al.	N/A	N/A
2006/0123436	12/2005	Tanaka et al.	N/A	N/A
2006/0135042	12/2005	Frost et al.	N/A	N/A
2006/0250755	12/2005	Tilton et al.	N/A	N/A
2006/0274501	12/2005	Miller	N/A	N/A
2007/0006599	12/2006	Kawamura et al.	N/A	N/A
2007/0025081	12/2006	Berlin et al.	N/A	N/A
2007/0029011	12/2006	Thatcher	N/A	N/A
2007/0034360	12/2006	Hall	N/A	N/A
2007/0199340	12/2006	Knight et al.	N/A	N/A
2007/0213000	12/2006	Day	N/A	N/A
2007/0227710	12/2006	Belady et al.	N/A	N/A
2007/0267741	12/2006	Attlesey et al.	N/A	N/A
2008/0002364	12/2007	Campbell et al.	N/A	N/A
2008/0017355	12/2007	Attlesey et al.	N/A	N/A
2008/0026509	12/2007	Campbell et al.	N/A	N/A
2008/0029250	12/2007	Carlson et al.	N/A	N/A
2008/0030945	12/2007	Mojaver et al.	N/A	N/A
2008/0055845	12/2007	Murakami et al.	N/A	N/A
2008/0158818	12/2007	Clidas et al.	N/A	N/A
2008/0174954	12/2007	Vangilder et al.	N/A	N/A
2008/0180908	12/2007	Wexler	N/A	N/A
2008/0196868	12/2007	Attlesey et al.	N/A	N/A
2008/0196870	12/2007	Attlesey et al.	N/A	N/A
2008/0209931	12/2007	Stevens	N/A	N/A
2008/0266726	12/2007	Murakami et al.	N/A	N/A
2008/0270572	12/2007	Belady et al.	N/A	N/A
2008/0273306	12/2007	Campbell et al.	N/A	N/A
2009/0168345	12/2008	Martini	N/A	N/A
2009/0219681	12/2008	Goth et al.	N/A	N/A
2009/0229194	12/2008	Armillas	N/A	N/A
2009/0251860	12/2008	Belady et al.	N/A	N/A
2009/0260777	12/2008	Attlesey	N/A	N/A
2009/0262455	12/2008	Merrow	N/A	N/A
2009/0295167	12/2008	Clidas et al.	N/A	N/A
2010/0027212	12/2009	Daunert Armillas	N/A	N/A
2010/0030267	12/2009	Winslow et al.	N/A	N/A
2010/0033070	12/2009	Cheng	N/A	N/A
2010/0061057	12/2009	Dersch et al.	N/A	N/A
2010/0103618	12/2009	Campbell et al.	N/A	N/A

2010/0139887	12/2009	Slessman	N/A	N/A
2010/0165565	12/2009	Hellriegal et al.	N/A	N/A
2010/0226094	12/2009	Attlesey et al.	N/A	N/A
2010/0246118	12/2009	Attlesey	N/A	N/A
2010/0263885	12/2009	Tuma	N/A	N/A
2010/0275441	12/2009	Rasmussen et al.	N/A	N/A
2010/0290190	12/2009	Chester et al.	N/A	N/A
2010/0302678	12/2009	Merrow	N/A	N/A
2010/0328889	12/2009	Campbell et al.	N/A	N/A
2011/0075353	12/2010	Attlesey et al.	N/A	N/A
2011/0103019	12/2010	Campbell et al.	N/A	N/A
2011/0120885	12/2010	Miller et al.	N/A	N/A
2011/0132579	12/2010	Best	165/104.31	H05K 7/20327
2011/0134604	12/2010	Attlesey	N/A	N/A
2011/0151765	12/2010	Chen et al.	N/A	N/A
2011/0157829	12/2010	Wormsbecher et al.	N/A	N/A
2011/0183051	12/2010	Skender	N/A	N/A
2011/0240281	12/2010	Avery	N/A	N/A
2011/0267775	12/2010	Vanderveen et al.	N/A	N/A
2012/0026691	12/2011	Campbell et al.	N/A	N/A
2012/0035773	12/2011	Stabinski et al.	N/A	N/A
2012/0123595	12/2011	Bower, III et al.	N/A	N/A
2012/0155027	12/2011	Broome et al.	N/A	N/A
2012/0171943	12/2011	Dunnavant	N/A	N/A
2012/0236487	12/2011	Wallace et al.	N/A	N/A
2012/0294737	12/2011	Singh et al.	N/A	N/A
2012/0300391	12/2011	Keisling et al.	N/A	N/A
2013/0021746	12/2012	Campbell et al.	N/A	N/A
2013/0025888	12/2012	Eckholm et al.	N/A	N/A
2013/0032217	12/2012	Pesek et al.	N/A	N/A
2013/0075066	12/2012	Reytblat	N/A	N/A
2013/0091814	12/2012	Smith	N/A	N/A
2013/0105120	12/2012	Campbell et al.	N/A	N/A
2013/0105139	12/2012	Campbell et al.	N/A	N/A
2013/0146273	12/2012	Chester et al.	N/A	N/A
2014/0060773	12/2013	Wajima	N/A	N/A
2014/0060799	12/2013	Eckberg et al.	N/A	N/A
2014/0085817	12/2013	Campbell et al.	N/A	N/A
2014/0085821	12/2013	Regimbal et al.	N/A	N/A
2014/0109610	12/2013	Wulf et al.	N/A	N/A
2014/0124167	12/2013	Campbell et al.	N/A	N/A
2014/0211412	12/2013	Best	N/A	N/A
2014/0216686	12/2013	Shelnutt et al.	N/A	N/A
2014/0216688	12/2013	Shelnutt et al.	N/A	N/A
2014/0218858	12/2013	Shelnutt et al.	N/A	N/A
2014/0301037	12/2013	Best	N/A	N/A
2014/0307384	12/2013	Best	N/A	N/A
2014/0362527	12/2013	Best	N/A	N/A

2015/0048950	12/2014	Zeighami et al.	N/A	N/A
2015/0062806	12/2014	Shelnutt et al.	N/A	N/A
2015/0138723	12/2014	Shedd et al.	N/A	N/A
2015/0181762	12/2014	Boyd et al.	N/A	N/A
2015/0195953	12/2014	Best et al.	N/A	N/A
2015/0245539	12/2014	Pelletier et al.	N/A	N/A
2015/0276292	12/2014	Son et al.	N/A	N/A
2015/0305209	12/2014	Inaba et al.	N/A	N/A
2015/0334880	12/2014	Best	N/A	N/A
2015/0359145	12/2014	Zhang et al.	N/A	N/A
2016/0234970	12/2015	Shelnutt et al.	N/A	N/A
2017/0142868	12/2016	Chen et al.	N/A	N/A
2017/0265328	12/2016	Sasaki et al.	N/A	N/A
2017/0265358	12/2016	Trees	N/A	N/A
2017/0303434	12/2016	Katsumata et al.	N/A	N/A
2017/0303443	12/2016	Inano et al.	N/A	N/A
2017/0332514	12/2016	Saito	N/A	N/A
2017/0359918	12/2016	Klaba et al.	N/A	N/A
2018/0020571	12/2017	Saito	N/A	N/A
2018/0020572	12/2017	Fujiwara et al.	N/A	N/A
2018/0288906	12/2017	Hopton et al.	N/A	N/A
2018/0368281	12/2017	Wang et al.	N/A	N/A
2019/0090383	12/2018	Tufty et al.	N/A	N/A
2019/0200482	12/2018	Boyd et al.	N/A	N/A
2019/0329973	12/2018	Chabot	N/A	N/A
2019/0344424	12/2018	Reynolds	N/A	N/A
2019/0387641	12/2018	Aoki et al.	N/A	N/A
2020/0093038	12/2019	Enright et al.	N/A	N/A
2020/0323108	12/2019	Bilan et al.	N/A	N/A
2021/0105910	12/2020	McCarthy	N/A	N/A
2021/0290430	12/2020	Kim	N/A	A61M 35/003
2021/0368656	12/2020	Heydari	N/A	N/A
2021/0382533	12/2020	Heydari	N/A	N/A
2021/0400834	12/2020	Shen et al.	N/A	N/A
2022/0220976	12/2021	Perevozchikov et al.	N/A	N/A
2022/0248564	12/2021	Moon	N/A	N/A
2023/0052701	12/2022	Chen et al.	N/A	N/A
2023/0091814	12/2022	Gordon et al.	N/A	N/A
2023/0164953	12/2022	Gregory et al.	N/A	N/A
2023/0209781	12/2022	Gao	700/281	H05K 7/20272
2023/0269900	12/2022	Kelly et al.	N/A	N/A
2023/0389231	12/2022	Chen	N/A	H05K 7/203

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

2009282170	12/2013	AU	N/A
101443724	12/2008	CN	N/A
109769381	12/2018	CN	N/A
109874275	12/2018	CN	N/A
111240448	12/2019	CN	N/A
113795118	12/2020	CN	N/A
114138084	12/2021	CN	N/A
115407848	12/2021	CN	N/A
82014500002000	12/2021	GB	N/A
2004319628	12/2003	JP	N/A
2004363308	12/2003	JP	N/A
2005015171	12/2004	JP	N/A
2011518395	12/2010	JP	N/A
2018019038	12/2017	JP	N/A
2019161201	12/2018	JP	N/A
2021519978	12/2020	JP	N/A
20150018884	12/2014	KR	N/A
2007023130	12/2006	WO	N/A
2007098078	12/2006	WO	N/A
2008027931	12/2007	WO	N/A
2008089322	12/2007	WO	N/A
2010019517	12/2009	WO	N/A
2013022805	12/2012	WO	N/A
2019061721	12/2018	WO	N/A
2020234600	12/2019	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion in PCT/US2024/035908, mailed Oct. 18, 2024, 9 pages. cited by applicant

International Preliminary Report on Patentability and Written Opinion, Application No. PCT/US2023/016342, mailed Oct. 10, 2024 (5 Pages). cited by applicant

U.S. Appl. No. 18/244,844; Non-Final Office Action issued Nov. 6, 2024; 65 pages. cited by applicant

European Patent Office, "Partial Supplementary European Search Report," issued in related European Patent Application No. 21892814.1, mailed Aug. 30, 2024 (15 pages). cited by applicant
Non-Final Office Action in U.S. Appl. No. 17/900,541, mailed Jun. 17, 2024, 8 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2023/026959, mailed Oct. 23, 2023, 8 pages. cited by applicant

Office Action for Chinese Application No. 200980131707.3, mailed Apr. 3, 2013, 3 pages. cited by applicant

Office Action for Chinese Application No. 200980131707.3, mailed Dec. 13, 2013, 4 pages. cited by applicant

International Search Report and Written Opinion in PCT/US2023/032438, mailed Dec. 27, 2023, 9 pages. cited by applicant

International Preliminary Report on Patentability in corresponding Application No. PCT/US2022/043817, mailed Mar. 28, 2024 (6 pages). cited by applicant

International Search Report and Written Opinion in corresponding Application No. PCT/US2024/010084, mailed May 1, 2024 (11 pages). cited by applicant

Office Action from U.S. Appl. No. 17/523,403, dated Apr. 23, 2024, 26 pages. cited by applicant

Applicant-Initiated Interview Summary from U.S. Appl. No. 13/057,881, dated Feb. 25, 2014, 4 pages. cited by applicant

Applicant-Initiated Interview Summary from U.S. Appl. No. 13/057,881, dated Jul. 1, 2014, 3 pages. cited by applicant

Applicant-Initiated Interview Summary from U.S. Appl. No. 13/057,881, dated Oct. 9, 2014, 3 pages. cited by applicant

Bakker A., "Modeling Flow Fields in Stirred Tanks," Reacting Flows—Lecture 7, 2006, 40 pages. cited by applicant

Communication pursuant to Article 94(3) EPC for European Patent Application No. 09807136.8, dated Nov. 26, 2018, 13 pages. cited by applicant

"Disruptive Technologies," SC|09, Disruptive Technologies Presentations, Retrieved from <http://sc09.supercomputing.org/index-pg=disrupttech.html>, Nov. 14-20, 2009, 2 pages. cited by applicant

"Disruptive Technologies," SC10 Disruptive Technologies, Retrieved from <http://sc10.supercomputing.org/index-pg=disrupttech.html>, Nov. 13-19, 2010, 8 pages. cited by applicant

Extended European Search Report for European Application No. 09807136.8, mailed Jun. 6, 2017, 12 Pages. cited by applicant

Extended European Search Report for European Application No. 12821634.8, mailed Jul. 15, 2015, 6 pages. cited by applicant

Final Office Action from U.S. Appl. No. 13/057,881, dated Jun. 16, 2015, 30 pages. cited by applicant

Final Office Action from U.S. Appl. No. 13/057,881, dated Nov. 22, 2013, 13 pages. cited by applicant

Final Office Action from U.S. Appl. No. 13/057,881, dated Nov. 3, 2014, 23 pages. cited by applicant

"GRC ICE Immersion Cooling Solutions Brochure," Retrieved from https://www.grcooling.com/wp-content/uploads/2018/06/GRC_ICE_Product_Brochure.pdf, Jun. 2018, 4 pages. cited by applicant

"GRC ICeraQ® Series 10 Launch Event Presentation," Retrieved from <https://digitalinfranetwork.com/videos/655447036-2/>, Dec. 27, 2021, 1 page. cited by applicant

"GRC® Releases the Next Generation of Data Center Liquid Immersion Cooling, Providing Increased Sustainability Capable of Delivering a PUE Yield of 1.03 and a Significant Reduction in Operating Costs," Retrieved <https://www.businesswire.com/news/home/20210518005857/en/GRC%C2%AE-Releases-the-Next-Generation-of-Data-Center-Liquid-Immersion-Cooling-Providing-Increased-Sustainability-Capable-of-Delivering-a-PUE-Yield-of-1.03-and-a-Significant-Reduction-in-Operating-Costs>, May 18, 2021, 3 pages. cited by applicant

"Green Revolution Cooling Named a Disruptive Technology of the Year at SC'10," Press Release, Nov. 1, 2010, 1 page. cited by applicant

"Green Revolution Cooling Named Disruptive at SC'09," Press Release, Aug. 3, 2009, 1 page. cited by applicant

Hernandez P., "Are Liquid-Cooled Servers Coming to a Data Center Near You?," CleanTech, Dec. 4, 2009, 2 pages. cited by applicant

Hernandez P., "Will IT Managers Warm to Liquid-Cooled Servers?," Green Revolution Cooling, Dec. 8, 2009, 3 pages. cited by applicant

"ICeraQ 10 Quad Duo-Specification Sheet," Retrieved from https://www.grcooling.com/wp-content/uploads/2018/06/GRC_Data_Sheet_ICeraQ_Quad-Duo-One.pdf, 2021, 2 pages. cited by applicant

Intel, "Intel® Core™ 2 Duo Processor on 65 nm Process for Embedded Applications," Thermal

Design Guide, Aug. 2007, pp. 1, 14, and 21. cited by applicant

Intel, "Intel® Pentium® 4 Processor on 90 nm Process," Datasheet, Feb. 2005, pp. 64-65. cited by applicant

"Intel Xeon Processor E7-4800/8800 v3 Product Families", May 2015, p. 46 (1 page). cited by applicant

International Preliminary Report on Patentability, Application No. PCT/US2014/033081, mailed Oct. 15, 2015, 7 pages. cited by applicant

International Preliminary Report on Patentability, Application No. PCT/US2014/037005, mailed Nov. 19, 2015, 12 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/US2009/053305, mailed Feb. 24, 2011, 19 pages. cited by applicant

International Preliminary Report on Patentability for International Application No. PCT/US2015/030618, mailed Nov. 24, 2016, 10 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2009/053305, mailed Oct. 14, 2009, 17 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US20115/030618, mailed Aug. 5, 2015, 12 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2012/049668, mailed Oct. 19, 2012, 10 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2014/017835, mailed Jun. 13, 2014, 9 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2014/033081, mailed Aug. 27, 2014, 9 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2014/033834, mailed Sep. 15, 2014, 16 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2014/037005, mailed Sep. 24, 2014, 18 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2015/43468, mailed Oct. 30, 2015, 11 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2021/058991, dated Mar. 3, 2022, 10 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2022/043817, mailed Jan. 10, 2023, 10 pages. cited by applicant

Lee S.Y., "Hydraulics and Mixing Evaluations for NT-21/41 Tanks," Revision 0, Oct. 2014, 32 pages. cited by applicant

Markoff J., "Data Centers' Power Use Less Than Was Expected," The New York Times, Jul. 31, 2011, 3 pages. cited by applicant

Miller R, "Data Center Used to Heat Swimming Pool," Data Center Knowledge, Retrieved from <https://www.datacenterknowledge.com/archives/2008/04/02/data-center-used-to-heat-swimming-pool>, Apr. 2, 2008, 2 pages. cited by applicant

Non-Final Office Action in U.S. Appl. No. 14/667,091, mailed May 22, 2015, 30 pages. cited by applicant

Non-final Office Action, U.S. Appl. No. 14/237,100, mailed Sep. 30, 2015, 21 pages. cited by applicant

Non-final Office Action, U.S. Appl. No. 14/245,978, mailed Jan. 13, 2016, 11 pages. cited by applicant

Non-final Office Action, U.S. Appl. No. 14/251,014, mailed Dec. 4, 2015, 13 pages. cited by applicant

Non-final Office Action, U.S. Appl. No. 14/271,386, mailed Dec. 17, 2015, 17 pages. cited by applicant

Notice of Acceptance for Australian Application No. 2009282170, mailed Nov. 14, 2014, 2 pages. cited by applicant

Office Action and Search Report for Chinese Application No. 200980131707.3, mailed Dec. 20, 2012, 17 pages. cited by applicant

Office Action for Canadian Application No. 2731994, mailed Dec. 3, 2015, 6 pages. cited by applicant

Office Action for Canadian Application No. 2731994, mailed Nov. 8, 2016, 4 pages. cited by applicant

Office Action for Chinese Application No. 200980131707.3, mailed Apr. 18, 2014, 7 pages. cited by applicant

International Search Report and Written Opinion for International Application No. PCT/US2023/016342, mailed Jul. 13, 2023, 8 pages. cited by applicant

Office Action for Chinese Application No. 200980131707.3, mailed Dec. 31, 2014, 10 pages. cited by applicant

Office Action for Chinese Application No. 200980131707.3, mailed Jul. 31, 2013, 8 pages. cited by applicant

Office Action for Chinese Application No. 201280049039.1, mailed Oct. 27, 2015, 33 pages. cited by applicant

Office Action from U.S. Appl. No. 13/057,881, dated Apr. 11, 2014, 16 pages. cited by applicant

Office Action from U.S. Appl. No. 13/057,881, dated Jan. 9, 2015, 3 pages. cited by applicant

Office Action from U.S. Appl. No. 13/057,881, dated Jun. 4, 2013, 13 pages. cited by applicant

Office Action from U.S. Appl. No. 13/057,881, dated Mar. 3, 2015, 20 pages. cited by applicant

Park J, "New Cooling Strategies for Greater Data Center Energy Efficiency," Facebook, Nov. 4, 2010, 4 pages. cited by applicant

Patent Examination Report No. 2 for Australian Application No. 2009282170, mailed Jun. 18, 2014, 4 pages. cited by applicant

Patent Examination Report No. 2 for Australian Application No. 2009282170, mailed Nov. 15, 2013, 4 pages. cited by applicant

Substantive Examination Adverse Report for Malaysian Application No. PI2011000494, mailed May 15, 2015, 3 pages. cited by applicant

Supplementary Examination Report for Singapore Application No. 11202000306X, mailed Nov. 18, 2022, 4 pages. cited by applicant

"The Green500 List—Nov. 2013," The Green 500, Nov. 20, 2013, 3 pages. cited by applicant

U.S. Appl. No. 13/057,881, filed Feb. 7, 2011, Christiaan Scott Best, 72 pages. cited by applicant

U.S. Appl. No. 14/338,013, filed Jul. 22, 2014, Christiaan Scott Best, 60 pages. cited by applicant

U.S. Appl. No. 14/338,020, filed Jul. 22, 2014, Christiaan Scott Best, 60 pages. cited by applicant

U.S. Appl. No. 14/338,026, filed Jul. 22, 2014, Christiaan Scott Best, 60 pages. cited by applicant

U.S. Appl. No. 14/338,035, filed Jul. 22, 2014, Christiaan Scott Best, 60 pages. cited by applicant

U.S. Appl. No. 14/667,091, filed Mar. 24, 2015, Christiaan Scott Best, 65 pages. cited by applicant

Written Opinion and Search Report for Singapore Application No. 201100595-6, mailed Apr. 16, 2012, 20 pages. cited by applicant

Written Opinion and Search Report for Singapore Application No. 201100595-6, mailed May 2, 2012, 21 pages. cited by applicant

Primary Examiner: Cao; Chun

Attorney, Agent or Firm: Greenberg Traurig, LLP

Background/Summary

BACKGROUND

(1) Immersion cooling systems may be used to cool power distribution components of computer systems, such as commercial computer servers, by submerging those components in a tank filled with a dielectric coolant (sometimes referred to simply as a fluid). Often, such immersion cooling systems may be used with multiple off-the-shelf smaller racks and use tanks that may require separate cooling systems for each rack.

(2) Although a single pump and heat exchanger may be used to cool multiple racks (i.e., a shared coolant distribution system), problems may arise in instances in which the multiple racks are desired to be cooled at different rates. In instances in which minor differences in coolant flow are used to vary the cooling rates of the racks, a difference in coolant levels in the various racks may be introduced. The difference in coolant levels may be a risk to the racks located in other tanks in instances in which the coolant levels raise too high or the coolant levels fall too low. In instances in which the coolant level is too high, there may be a risk that the coolant level may overflow the tank containing the rack. In instances in which the coolant level is too low, there may be the risk of exposing parts or all of the computer system to air, which can cause overheating due to insufficient coolant and cooling. In addition, in instances in which coolant levels fall to too low, there may be the risk of introducing air into the coolant circuit, which may damage pumps that circulate the coolant. While the coolant may be pumped out of the bottom of the tanks to avoid air intake, a subsequent leak or failure at a valve and/or duct located near the bottom of the tank may result in a complete draining of the tank. This in turn may again run the risk of exposing parts or all of the computer system to air, which can cause overheating due to insufficient coolant and cooling.

SUMMARY

(3) Various aspects of the present disclosure include methods, systems, and devices for controlling coolant levels in a plurality of component cooling tanks sharing a unified coolant distribution system. Various aspects may include first coolant level regulator components of a first component cooling tank, second coolant level regulator components of a second component cooling tank, and a processor. The first coolant level regulator components of the first component cooling tank may include a first temperature sensor configured to measure a temperature of the coolant in a first reservoir, a first coolant release valve configured to regulate the release of coolant from the first reservoir, and a first coolant level detector configured to detect a level of the coolant in the first reservoir. The second coolant level regulator components of the second component cooling tank may include a second temperature sensor configured to measure a temperature of the coolant in a second reservoir, a second coolant release valve configured to regulate the release of coolant from the second reservoir, and a second coolant level detector configured to detect a level of the coolant in the second reservoir. The processor may be configured to receive the temperature measurements from the first temperature sensor and the second temperature sensor. The processor may also be configured to signal at least one target coolant level update for adjusting a level of the coolant in the first reservoir by the first coolant release valve. The at least one target coolant level update may be determined based on a temperature average of the received temperature measurements from the first temperature sensor and the second temperature sensor. The first coolant release valve may be configured to adjust a release of the coolant from the first reservoir in response to comparing the detected level of the coolant in the first reservoir to a target coolant level indicated by the at least one target coolant level update.

(4) The first coolant level detector may include an ultrasonic fluid level sensor. The first coolant level regulator components may include a first coolant level controller configured to receive the signaled at least one target coolant level update. The first coolant level controller may be

configured to activate the first coolant release valve in response to receiving the at least one target coolant level update. The first coolant level controller may be configured to operate independent of the processor configured to receive the temperature measurements from the first temperature sensor and the second temperature sensor. The signaled at least one target coolant level update from the processor may be configured to directly control at least the first coolant release valve for regulating the release of coolant associated therewith. The signaled at least one target coolant level update from the processor may be determined by converting the temperature average of the received temperature measurements to a density average of the coolant in the first reservoir and the second reservoir.

(5) In some embodiments, the first coolant release valve may include a linear actuator configured to control a state of a coolant gate regulating the release of coolant. In some embodiments, the first coolant release valve may include a rotary actuator configured to control a state of a coolant gate regulating the release of coolant.

(6) In some embodiments, the processor may be further configured to signal the at least one target coolant level update for adjusting a level of the coolant in at least one of the second reservoir by at least the second coolant release valve, wherein the at least one target coolant level update is determined based on the temperature average of the received temperature measurements from the first temperature sensor and the second temperature sensor.

(7) In some embodiments, the system may further include third coolant level regulator components of a third component cooling tank and fourth coolant level regulator components of a fourth component cooling tank. The third coolant level regulator components of the third component cooling tank may include a third temperature sensor configured to measure a temperature of the coolant in a third reservoir, a third coolant release valve configured to regulate the release of coolant from the third reservoir, and a third coolant level detector configured to detect a level of the coolant in the third reservoir. The fourth coolant level regulator components of a fourth component cooling tank may include a fourth temperature sensor configured to measure a temperature of the coolant in a fourth reservoir, a fourth coolant release valve configured to regulate the release of coolant from the fourth reservoir, and a fourth coolant level detector configured to detect a level of the coolant in the fourth reservoir. The processor may be configured to receive the temperature measurements from the third temperature sensor and the fourth temperature sensor. The at least one target coolant level update signaled by the processor may be configured to adjust a level of the coolant in at least one of the third reservoir or the fourth reservoir by the third coolant release valve or the fourth coolant release valve, respectively.

(8) In some embodiments, the first reservoir and the second reservoir may be configured to hold an overflow of the coolant from the first component cooling tank and the second component cooling tank, respectively. The first reservoir and the second reservoir may each be the largest coolant reservoirs of the first component cooling tank and the second component cooling tank, respectively.

(9) Various embodiments include a method for controlling coolant levels in two or more component cooling tanks sharing a unified coolant distribution system. The method includes receiving, at a processor of the unified coolant distribution system, temperature measurements of a coolant in a first reservoir and a second reservoir of the two or more component cooling tanks. Also, the method includes receiving, at the processor of the unified coolant distribution system, a coolant level measurement of the coolant in the first reservoir. Additionally, the method includes signaling, by the processor to a coolant level regulator component of each of the two or more component cooling tanks, a target coolant level update configured to equalize levels of the coolant in the first reservoir and the second reservoir to a target coolant level, wherein the target coolant level is determined based on the received coolant level measurement and a temperature average of the received temperature measurements of the coolant in the first reservoir and the second reservoir.

(10) In some embodiments, the target coolant level update signaled by the processor may be determined by converting the temperature average of the received temperature measurements to a

density average of the coolant in the first reservoir and the second reservoir.

(11) Signaling the target coolant level update may include signaling a first coolant level controller configured to control a first coolant release valve that regulates the release of coolant from the first reservoir and a second coolant level controller configured to control a second coolant release valve that regulates the release of coolant from the first reservoir.

(12) Signaling the target coolant level update may include directly activating at least one of a first coolant release valve that regulates the release of coolant from the first reservoir or a second coolant release valve that regulates the release of coolant from the first reservoir.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and constitute part of this specification, illustrate exemplary embodiments of the claims and together with the general description given above and the detailed description given below, serve to explain the features of the claims.

(2) FIG. 1A is a schematic view of a single-phase cooling distribution unit used to cool two component cooling tanks in accordance with various embodiments.

(3) FIG. 1B is a schematic view of a single-phase cooling distribution unit used to cool four component cooling tanks in accordance with various embodiments.

(4) FIG. 2A is a perspective view of a system schematic for controlling coolant levels in component cooling tanks sharing a unified coolant distribution system in accordance with various embodiments.

(5) FIG. 2B is a schematic relief view of a coolant level detector suitable for use with various embodiments.

(6) FIGS. 3A and 3B are vertical cross-sectional views of a component cooling tank including a reservoir with a coolant release valve in closed and open positions, respectively, in accordance with various embodiments.

(7) FIG. 4A is a perspective relief view of a linear coolant release valve in a reservoir in accordance with various embodiments.

(8) FIG. 4B is a perspective relief view of a rotary coolant release valve in a reservoir in accordance with various embodiments.

(9) FIG. 5 is a process flow diagram of a method for controlling coolant levels in two or more component cooling tanks sharing a unified coolant distribution system in accordance with various embodiments.

DETAILED DESCRIPTION

(10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting.

(11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(12) Various embodiments will be described in detail with reference to the accompanying drawings. Wherever possible, the same reference numbers will be used throughout the drawings to refer to the

same or like parts. References made to particular examples and implementations are for illustrative purposes and are not intended to limit the scope of the claims.

(13) In various embodiments, adjustments to the coolant level in each of a plurality of coolant tanks that share a volume of circulated coolant may be made to one, some or all of the plurality of the coolant tanks based on an average temperature of the coolant across the plurality of coolant tanks. The measure of the average temperature may be used to determine a current volume of the bulk fluid of the shared volume of circulated coolant across the plurality of coolant tanks. The measure of the average temperature may in turn be used to adjust an amount of coolant (i.e., volume of coolant) maintained in the coolant tanks or at least overflow, buffer, and/or outflow reservoirs thereof. To control coolant levels across the plurality of component cooling tanks, various embodiments may adjust coolant levels in each of the plurality of coolant tanks themselves and/or the reservoirs thereof. Thus, while various embodiments herein are described as adjusting the volume of the “reservoirs,” it should be understood that the coolant level regulator components described herein may be used in the reservoirs and/or directly in the main reservoirs of each of the plurality of coolant tanks that hold the components being cooled. In this way, as used herein the term “reservoir” may refer to either main component cooling tank reservoirs, outflow reservoirs, buffer reservoirs, and/or overflow reservoirs.

(14) The manifolds that allow coolant into multiple reservoirs typically provide uneven coolant flow into the different reservoirs. Even if the reservoir inlets were equipped with reservoir inlet control valves, which they are not typically equipped with, the inflow going to different reservoirs may vary. Also, while adjustable valves may be added to each of the inlet ports feeding coolant into the separate reservoirs, often space is more limited in or around the manifold. Thus, various embodiments add dynamically regulated valves at an exit port (i.e., an outlet) for the coolant in each reservoir. Without separately regulating the outlet ports of the manifold or inlet ports of the reservoirs, various embodiments dynamically regulate the outlet ports of the reservoirs.

(15) In accordance with various embodiments, the height of the volume of coolant maintained in each of the plurality of reservoirs may be adjusted based on the measured average temperature. These adjustments may encourage an equal height of coolant (i.e., fluid) across the plurality of reservoirs. The system may attempt to keep all the reservoirs at the same coolant level in order to maintain an even hydraulic head for the coolant exiting all the reservoirs, which encourages the outflow of coolant from the reservoirs to feed into the suction return evenly.

(16) In accordance with various embodiments, the coolant height and temperature measurements may be communicated to a system controller. The system controller may be configured to make determinations regarding the coolant height and temperature, as well as direct the reservoir controller to change coolant levels if necessary. For example, using a Modbus remote terminal unit (RTU), programmable logic controller (PLC) controller, or a controller area network (CAN), multiple reservoir controllers may communicate with the shared controller. In this way, the shared controller may determine a current average coolant height for a plurality of reservoirs and compare that to a target average coolant height for the plurality of reservoirs. In instances in which the current coolant height in any or all of the plurality of reservoirs is above or below the target average coolant height, flow adjustments may be made to bring coolant height in each reservoir to the target average. The reservoir controller(s) may thus adjust the coolant levels in response to receiving instructions to do so from the shared controller.

(17) The coolant level reflects not only the mass of coolant, but also the coolant temperature since the volume of coolant for a given mass will change as a function of coolant bulk temperature. Thus, a low coolant level for coolant that is at a relatively high temperature may be more alarming than the same low coolant level for coolant that is at a relatively low temperature. In response to detecting a temperature change across the reservoirs, various embodiments may change a target average reservoir coolant height accordingly.

(18) In accordance with various embodiments, a central or unified control system may evaluate

various elements of the systems parameters to report abnormal situation and may autonomously adjust things to a safe coolant level in the event of communication loss between control elements of the more than one return reservoir monitoring and control elements. In addition, or alternatively, individual coolant level controllers for each component cooling tank may operate autonomously, which may provide a backup or failsafe upon failure of the main system controller. Each of the plurality of reservoirs may communicate their respective coolant level and temperature to a shared controller for determining a current average coolant level, which may be used as a control point. That control point may be used to determine a new target average coolant height, which considers the current coolant temperature. That new target average coolant height may be used to adjust coolant heights in all the reservoirs accordingly.

(19) FIG. 1A illustrates a single-phase cooling distribution system **100** suitable for various embodiments. The single-phase cooling distribution system **100** includes a single-phase cooling distribution unit **20** configured to remove thermal energy from two or more component cooling tanks **110**, which may be open racks. In particular, FIG. 1A includes two component cooling tanks **110** being cooled by the single-phase cooling distribution unit **20**. The single-phase cooling distribution unit **20** may include a coolant pump that circulates the coolant through a coolant-to-water heat exchanger. The water passing through the heat exchanger may be configured to absorb heat from the heated coolant and dissipate the absorbed heat in a cooling tower **30**, such as an evaporative cooling tower, dry cooler, existing chilled water loop, or other cooling system. The heated coolant absorbs heat from electronic components disposed within the component cooling tanks **110**. The heated coolant may be circulated to the coolant-to-water heat exchanger included in the single-phase cooling distribution unit **20**. Cooled coolant may be circulated from the coolant-to-water heat exchanger included in the single-phase cooling distribution unit **20** back into the component cooling tanks **110**. Alternatively, an open loop cooling water supply and disposal system may be used. The single-phase cooling distribution unit **20** may circulate coolant as a coolant input flow rate C.sub.IN-1, C.sub.IN-2 through each of the two component cooling tanks **110** to dissipate the thermal energy generated by electronic equipment disposed therein (i.e., components). The individual component cooling tanks **110** are considered “open” or referred to as “open racks,” if they have a liquid to air interface with the air at atmospheric pressure. The component cooling tanks **110** may have lids or covers but need not be sealed closed to allow the air to maintain atmospheric pressure. The individual component cooling tanks **110** may be configured to hold computing devices that generate heat. The component cooling tanks **110** are configured to receive and hold the circulating single-phase coolant used to dissipate that heat. Heated coolant may exit the component cooling tanks **110** as a coolant exit flow rate (C.sub.OUT-1, C.sub.OUT-2), which be recirculated back to the single-phase cooling distribution unit **20** for the cycle to repeat. Although FIG. 1 shows two (2) component cooling tanks **110**, alternative embodiments may have three or more component cooling tanks **110**.

(20) As the temperature of the coolant heats up, the coolant expands, which means the volume of coolant increases and the density thereby decreases. The opposite occurs when the coolant cools down. Since each of the component cooling tanks may contain different electronic components, a different number of electronic components, and/or electronic components that are operating at different levels causing different heat generation. Thus, an uneven generation of heat may occur among different component cooling tanks **110**. As a result, the volume of coolant retained within the individual component cooling tanks **110** may also vary. This also results in all or many of the component cooling tanks **110** potentially having different coolant exit flow rates C.sub.OUT-1, C.sub.OUT-2. Such disparities between component cooling tanks **110** may cause a potential for unequal flow between parallel open component cooling tanks **110**. This unequal flow may create the possibility of component cooling tanks **110** having unequal coolant heights, particularly between their reservoirs when included. A reservoir of a component cooling tank may have the coolant level contained therein drop below a level of an outlet duct. This may in-turn cause air to

be pulled into the pipes and pump(s) circulating the coolant. In some cases, the imbalance may create unusually high coolant levels in one or more of the component cooling tanks **110** or the reservoirs, creating the risk of overflow and coolant being lost from the containment system.

(21) The thermal expansion/contraction of coolant as it cycles between minimum and maximum temperatures further complicates the matter. Reservoirs of the component cooling tanks **110** may be configured to buffer the volumetric change of dielectric coolant fluid over normal temperature ranges. However, as the volume and therefore level of coolant changes, due to thermal temperature changes, a simple mechanical float mechanism cannot compensate for the phenomena.

(22) Various embodiments include a system for controlling coolant levels in two or more component cooling tanks **110** sharing the single-phase cooling distribution system **100**. The system for controlling coolant levels may include coolant level regulator components **145** in each of the component cooling tanks **110**, as well as a controller **120** communicatively coupled to the coolant level regulator components **145**. The coolant level regulator components **145** may include a temperature sensor **140** and a coolant release valve **160**. Each temperature sensor **140** may be configured to measure a temperature of the coolant in a reservoir of the respective component cooling tanks **110**. Additionally, or alternatively, the temperature sensor **140** may be configured to measure air temperature, liquid temperature, or the temperature of solid matter. For example, the temperature sensor **140** may include a thermistor, resistance temperature detector (RTD), thermocouple, temperature probe, etc. Also, each coolant release valve **160** may be configured to regulate the release of coolant from the reservoir of the respective component cooling tanks **110**. A processor of the controller **120** may be configured to receive temperature measurements from the temperature sensors **140**. Also, based on the received temperature measurements, a processor of the controller **120** may determine a target coolant level for the reservoirs, which may require adjusting levels of the coolant in at least one of the reservoirs by the corresponding coolant release valves **160**. The controller **120** may signal the determined target coolant level to one or both of the coolant release valves **160**. Such a signal from the controller **120** may take the form of a target coolant level update. The target coolant level update may be configured to cause one or both of the coolant release valves **160** to adjust the coolant levels in their respective reservoirs.

(23) Additionally, each of the coolant level regulator components **145** may include a coolant level detector. A coolant level detector may be configured to detect a level of the coolant in the reservoir, which may be useful to determine whether to adjust a release of the coolant from the reservoir.

(24) As a further alternative and/or addition, each of the coolant release valves **160** may be managed by a coolant level controller, which may be a further component of the coolant level regulator components **145**. In this way, the controller **120** may signal the coolant level controller, which may in-turn adjust its corresponding coolant release valve **160**, if necessary.

(25) Various embodiments may use multiple immersion cooling rack assemblies **110** in parallel to reduce the cost per space of cooling. For example: four immersion cooling rack assemblies **110** of approximately 50 U may be connected to a single pump. While some datacenters use the same information technology load in each area or immersion cooling rack assembly **110**, some collocation facilities may have different loads from one immersion cooling rack to another. Customer may only require a single rack of space, which needs far fewer resources than a customer using multiple racks.

(26) FIG. 1B illustrates another single-phase cooling distribution system **101** suitable for various embodiments. The single-phase cooling distribution system **101** includes a single-phase cooling distribution unit **20** configured to remove thermal energy from four component cooling tanks **110**, similar to those components described above with regard to FIG. 1A. The single-phase cooling distribution unit **20** may circulate coolant at a coolant input flow rate C.sub.IN-1, C.sub.IN-2, C.sub.IN-3, C.sub.IN-4 through each of the four component cooling tanks **110** to dissipate the thermal energy generated by electronic equipment disposed therein. In addition, the component cooling tanks **110** may potentially have different coolant exit flow rates C.sub.OUT-1, C.sub.OUT-

2, C.sub.OUT-3, C.sub.OUT-4. Although FIG. 1B shows four (4) component cooling tanks **110**, alternative embodiments may have three or more than four component cooling tanks **110**. Like the single-phase cooling distribution system **100** described above with regard to FIG. 1A, the single-phase cooling distribution system **101** may include a system for controlling coolant levels, which may include similar coolant level regulator components **145** in each of the component cooling tanks **110**, as well as the controller **120** communicatively coupled to the coolant level regulator components **145**. Also, the coolant level regulator components **145** may include the temperature sensor **140** and the coolant release valve **160**, which the controller **120** may use to determine target coolant level updates for adjusting levels of the coolant in at least one of the reservoirs in the component cooling tanks **110**.

(27) FIG. 2A illustrates various aspects of a system **200** for controlling coolant levels in two component cooling tanks (e.g., **110** in FIG. 1A) sharing a unified coolant distribution system in accordance with various embodiments. The system **200** may include the controller **120**, which is communicatively coupled to coolant level regulator components, such as the temperature sensors **140-1**, **140-2** and the coolant release valves **160-1**, **160-2**. The controller **120** may have a wired and/or wireless connection directly to the coolant level regulator components (e.g., the temperature sensors **140-1**, **140-2** and the coolant release valves **160-1**, **160-2**) in each reservoir **130-1**, **130-2**. For example, the controller **120** may use a communication BUS **125** configured to communicatively couple the controller **120** to the coolant level regulator components. Alternatively, the controller **120** may have a wired and/or wireless connection to a coolant level controller **150-1**, **150-2**, which may be coupled to one or more of the coolant level regulator components.

(28) The reservoirs **130-1**, **130-2** may be configured to hold coolant **50** therein. In various embodiments, the reservoirs **130-1**, **130-2** may be configured to operate as overflow reservoirs, receiving overflow coolant from larger holding tanks, such as ones holding electronic components being cooled. The overflow coolant may flow into each of the reservoirs **130-1**, **130-2** through an open top **132-1**, **132-2** or other aperture at different overflow input flow rates O.sub.IN-1, O.sub.IN-2. In some embodiments, the reservoirs **130-1**, **130-2** may be configured as an outflow reservoir configured to collect coolant prior to it exiting through an outlet duct **138-1**, **138-2**. The reservoirs **130-1**, **130-2** may have extra headspace (i.e., volume) to buffer volumetric changes in coolant due to temperature changes. In addition, the reservoirs **130-1**, **130-2** may potentially have different coolant exit flow rates C.sub.OUT-1, C.sub.OUT-2. Alternatively, each reservoir **130-1**, **130-2** may form the main component holding reservoir of that respective component cooling tank (i.e., a tank without an overflow reservoir). In this way, the overflow input flow rate O.sub.IN-1, O.sub.IN-2 would be the equivalent to the coolant input flow rates C.sub.IN-1, C.sub.IN-2 described above with regard to FIG. 1A. Although the system **200** includes only two reservoirs **130-1**, **130-2**, additional reservoirs and/or component cooling tanks may be included.

(29) At room temperature or some other setup temperature, an initial coolant height H.sub.R-init of the surface level **52-1**, **52-2** of the coolant **50** in each reservoir **130-1**, **130-2** may be known, as well as the corresponding initial temperature T.sub.init (e.g., room temperature) and initial density of the coolant **50**. Similarly, since the dimensions of each reservoir **130-1**, **130-2** are known, a horizontal cross-sectional area A.sub.R of each reservoir **130-1**, **130-2** (or an average thereof) may be used in conjunction with the initial coolant height H.sub.R-init to determine an initial coolant volume V.sub.init. Over time, as the electronic components being cooled generate heat and such heat is transferred to the coolant, the temperature of the coolant **50** may increase. Also, as the temperature of the coolant **50** increases, fluid expansion may cause the coolant levels H.sub.R-1, H.sub.R-2 to rise in each of the reservoirs **130-1**, **130-2**. In addition, the temperature of the coolant **50** in different reservoirs **130-1**, **130-2** may vary since there may be variations in the amount of heat generated by the electronic components in different component cooling tanks. Thus, the coolant level rise may also vary from tank to tank.

(30) In accordance with various embodiments, each reservoir **130-1**, **130-2** may include a set of coolant level regulator components, which are configured to control the coolant levels H.sub.R-1, H.sub.R-2 in each of the reservoirs **130-1**, **130-2**. In this way, a first reservoir **130-1** (e.g., associated with a first component cooling tank) may include first coolant level regulator components. Similarly, a second reservoir **130-2** (e.g., associated with a second component cooling tank) may include second coolant level regulator components. As described above, the coolant level regulator components may include a temperature sensor **140** and a coolant release valve **160-1**, **160-2**. Each temperature sensor **140-1**, **140-2** may be configured to measure a temperature of the coolant **50** in one of the reservoirs **130-1**, **130-2**. Also, each coolant release valve **160-1**, **160-2** may be configured to regulate the release of coolant from one of the reservoirs **130-1**, **130-2** through the outlet ducts **138-1**, **138-2**.

(31) A processor of the controller **120** may receive a first temperature measurement T.sub.1 from a first temperature sensor **140-1** of the first reservoir **130-1** and receive a second temperature measurement T.sub.2 from a second temperature sensor **140-2** of the second reservoir **130-2**. Using the received temperature measurements T.sub.1, T.sub.2, the processor may determine a current average reservoir temperature T.sub.Avg. (e.g., (T.sub.1+T.sub.2)/2=T.sub.Avg). In a system with more than two reservoirs, temperature measurements from the additional reservoirs may be included in the temperature average calculation.

(32) The coolant **50** used in the reservoirs **130-1**, **130-2** (e.g., a dielectric coolant) may have well known properties, such as density ρ , which may vary with temperature. As noted above, an initial coolant temperature T.sub.init may be known, such as an ambient temperature (i.e., room temperature). In addition, since the density variations based on temperature are generally known, a correlation between temperature and density may be maintained in a lookup table, such as in memory of the controller **120** and/or one or more of the coolant level controllers **150-1**, **150-2**. Thus, the initial coolant temperature T.sub.init will have a corresponding initial coolant density ρ .sub.init. Also, as coolant temperatures change, the processor of the controller **120**, may use the lookup table to convert (i.e., translate) an average of the current measured temperatures (i.e., a temperature average) into an estimation of current average coolant density (i.e., a density average). For example, an average coolant temperature T.sub.Avg may be translated into an average coolant density ρ .sub.Avg.

(33) Based on temperatures measured by the temperature sensors **140-1**, **140-2** and the corresponding coolant density thereof, the controller **120** may determine a target coolant height H.sub.T, which may correspond to an optimal coolant level that should be maintained across all the reservoirs **130-1**, **130-2**. The processor may then use the coolant level regulator components to ensure or at least encourage the reservoir coolant levels get adjusted to or maintained at the target coolant height H.sub.T. For example, the processor may use the following equation to determine a target coolant height H.sub.T:

$$(34) \quad H_T = V_{init} * (\rho_{init} / \rho_{avg} - 1) / A_R + H_{R-init} \quad (1)$$

In equation (1), H.sub.T represents the target coolant height, V.sub.init represents the initial coolant volume (e.g., at setup), ρ .sub.init represents the initial coolant density, ρ .sub.avg represents an average coolant density based on the average measured coolant temperature (e.g., T.sub.Avg), A.sub.R represents the horizontal cross-sectional area of each reservoir **130-1**, **130-2**, and H.sub.R-init represents the initial coolant height in the reservoirs **130-1**, **130-2**. Alternatively, the processor may use a lookup table to directly determine the target coolant height H.sub.T based on the average measured coolant temperature T.sub.Avg.

(35) Once the target coolant height H.sub.T is determined, the controller **120** may initiate operations to control the coolant release valves **160-1**, **160-2** for adjusting coolant levels in the reservoirs **130-1**, **130-2**. For example, the controller **120** may signal a target coolant level update. The target coolant level update may be based on the determined target coolant height H.sub.T, for

adjusting levels of the coolant **50** in at least one of the first reservoir **130-1** or the second reservoir **130-2**. The target coolant level update may be a control signal transmitted directly to one or both of the first coolant release valve **160-1** and/or the second coolant release valve **160-2**. The first coolant release valve **160-1** and the second coolant release valve **160-2** may be adjustable, which in-turn adjusts a release of the coolant from the first reservoir **130-1** and the second reservoir **130-2**, respectively. The control signal may selectively open or close the first and/or second coolant release valve **160-1**, **160-2** to increase, maintain, or decrease the coolant levels therein.

Alternatively, each reservoir **130-1**, **130-2** may have an independent microcontroller, such as the coolant level controllers **150-1**, **150-2**, that provide proportional integral derivative (PID) closed loop control to adjust coolant exit flow rates C.sub.OUT-1, C.sub.OUT-2 of specific reservoirs **130-1**, **130-2** for adjusting and/or maintaining target coolant levels (e.g., H.sub.T).

(36) In various embodiments, the coolant level regulator components of the system **200** may also include one or more coolant level detectors configured to detect a level of the coolant **50** in the reservoirs **130-1**, **130-2**. For example, a first coolant level detector **170-1** may be disposed in or on the first reservoir **130-1**. Similarly, a second coolant level detector **170-2** may be disposed in or on the second reservoir **130-2**. One or both of the first coolant level detector **170-1** and the second coolant level detector **170-2** may be an ultrasonic fluid level meter, which is configured to provide precise fluid level measurements. FIG. 2B illustrates an ultrasonic coolant level detector **170-n** (i.e., **170-1** and/or **170-2**) suitable for use with various embodiments. The ultrasonic coolant level meter **170-n** may be a contactless measurement device that transmits (Tx) and receives (Rx) high frequency acoustic waves to measure the distance *d* to the surface **52-n** of the coolant **50**.

(37) Alternatively, the first coolant level detector **170-1** and the second coolant level detector **170-2** may be capacitance sensors, which may use probes that detect when the dielectric coolant level changes (i.e., rises or recedes). As the coolant level changes, the capacitance will change and provide an indication of the coolant level. As a further alternative, the first coolant level detector **170-1** and the second coolant level detector **170-2** may be pressure sensors configured to measure coolant level. As a further alternative, a float-type fluid level meter may be included, which tends to provide less accurate measurements, but may save costs. Other sensors may be used to alternatively, or additionally, measure the coolant levels H.sub.R-1, H.sub.R-2 in the reservoirs **130-1**, **130-2**.

(38) The first coolant level detector **170-1** and the second coolant level detector **170-2** may be configured to measure and report the coolant levels H.sub.R-1, H.sub.R-2 to the controller **120** and/or their respective coolant level controllers **150-1**, **150-2**, if included. The controller **120** and/or the coolant level controllers **150-1**, **150-2** may compare the current coolant levels H.sub.R-1, H.sub.R-2 to those indicated in the target coolant level update (e.g., H.sub.T). In instance in which any of the current coolant levels H.sub.R-1, H.sub.R-2 are above the target coolant level H.sub.T indicated by the target coolant level update, then the controller **120** or the respective coolant level controller **150-1**, **150-2** may either open or further open the first and/or second coolant release valve(s) **160-1**, **160-2** to increase the coolant exit flow rates C.sub.OUT-1, C.sub.OUT-2.

(39) FIGS. 3A and 3B illustrate a system **300** with a component cooling tank **110** that includes a main component reservoir **113** and an outflow reservoir **330** with coolant level regulator components for controlling coolant levels in the outflow reservoir **330** in accordance with various embodiments. The outflow reservoir **330** may include the coolant release valve **160**. FIG. 3A illustrates the coolant release valve **160** in a closed position and FIG. 3B illustrates the coolant release valve **160** in an open position. In various embodiments, the fully closed position of the coolant release valve **160** (as shown in FIG. 3A) may not completely close-off or prevent coolant fluid from flowing out the outlet duct **138** of the outflow reservoir **330**.

(40) The main component reservoir **113** may be configured to hold electronic components and includes a weir **115** that is configured to allow spill-over of coolant **40** from the main component reservoir **113** to the outflow reservoir **330**. In this way, in instances in which a top surface **41** of the

coolant **40** reaches the level of the weir **115**, any additional coolant **40** will spill over the weir **115** into the outflow reservoir **330**. The outflow reservoir **330** may be similar to the reservoirs described herein with regard to other embodiments (e.g., reservoir **130**).

(41) In addition to or as an alternative to the weir **115**, a simple partition wall or siphon ports may be included at the top of a wall between the main component reservoir **113** and outflow reservoir **330** instead of or in addition to the weir **115**. As a further alternative, regulated siphon ports may be included almost anywhere between the main component reservoir **113** and the outflow reservoir **330** to transfer excess coolant **40** therebetween.

(42) The outflow reservoir **330** may include coolant level regulator components, as described above, including the temperature sensor **140** and the coolant release valve **160**. The temperature sensor **140** is configured to measure the temperature of the coolant **50** in the outflow reservoir **330**. The coolant release valve **160** may be configured to regulate the release of coolant **50** from the outflow reservoir **330**. In addition, the coolant level regulator components may include the coolant level controller **150**, which may more directly control the coolant release valve **160** and receive measurements from the temperature sensor **140**. Further, the coolant level regulator components may include the coolant level detector **170**, which may be configured to detect a coolant level of a top surface **51** of the coolant **50** in the outflow reservoir **330**.

(43) The outflow reservoir **330** may be a separate tank from the component coolant tank **110** that forms the main component reservoir **113**. The outflow reservoir **330** is configured to receive overflow coolant from the main component reservoir **113**. The weir **115** may extend along an upper extent of the barrier separating the component coolant tank **110** from the overflow reservoir **330**. Also, that barrier between the two tanks **110**, **330** may be lower than the other walls of the component coolant tank **110**. The weir **115** may be formed as a flat horizontal strip, configured to allow excess coolant **40** to spill out from the component coolant tank **110**, over the weir **115**, and into the outflow reservoir **330**. In various embodiments, the weir **115** may extend from one side of the component coolant tank **110** to the other. In other embodiments, the weir **115** may only extend across a portion of the component coolant tank **110**. In other embodiments, more than one weir **115** may be provided, each extending across different portions of the component coolant tank **110**. In this manner, a weir **115** may be disposed on any and all edges of the component coolant tank **110** so that the component coolant tank **110** has one or more outflow reservoirs **330** around some or all of the perimeter of the component coolant tank **110**.

(44) FIGS. **4A** and **4B** illustrate different types of coolant release valves **160A**, **160B** for component coolant tanks in accordance with various embodiments. In particular, FIG. **4A** illustrates a linear coolant release valve **160A** and FIG. **4B** illustrates a rotary coolant release valve **160B**. The coolant release valves **160A**, **160B** may be disposed at the bottom of the reservoir **130**, with a whirlpool shield **175** disposed between each of the coolant release valves **160A**, **160B** and their respective outlet duct **138**. It may be advantageous to position the outlet duct **138** as low as possible within the reservoir **130** to maximize variance volume. Variance volume may be defined by the difference in volume of fluid in the collection zone between max and min levels.

(45) The whirlpool shield **175** may ensure only coolant **50**, and not air, is suctioned through the outlet duct **138**. The intake or suctioning of air into the outlet duct **138** may damage a pump (not shown) that is used to circulate the coolant **50**. The whirlpool shield **175** may be mounted inside the reservoir **130** above the outlet duct **138** for the coolant **50** to exit the reservoir **130**. A first end of the whirlpool shield **175** may be attached to a side wall of the reservoir **130**, such as the side wall including the outlet duct **138**. The whirlpool shield **175** may extend away from the first end (i.e., and the outlet duct **138**) toward a second end disposed further from the outlet duct **138** than the first end. The second end of the whirlpool shield **175** may be attached to the coolant release valve **160A**, **160B**. Also, the whirlpool shield **175** may extend downward at an angle (i.e., with a slope) such that the second end of the whirlpool shield **175** is vertically lower than the first end of the whirlpool shield **175**. Alternatively, the whirlpool shield **175** may be formed to have an L-shape,

extending away from the outlet **138**, toward the central part of the reservoir **130**, and then bending downward at a remote end thereof. Including the whirlpool shield **175** may lower the minimum coolant level needed to be maintained in the reservoir **130** before air gets sucked into the outlet duct **138**. In addition, the whirlpool shield **175** may prevent air bubbles caused by coolant **50** flowing over the weir **115** into the reservoir **130** from entering the outlet duct **138**. In other words, the whirlpool shield **175** may ensure only fluid is expelled from the bottom of the collection zone. Also, the whirlpool shield **175** may prevent whirlpool flows inside the reservoir **130**, particularly right next to the outlet duct **138**. For example, with the whirlpool shield **175** mounted immediately above a 2.5" diameter outlet duct aperture, the minimum fluid height may be lowered by inches, such as ½" from the bottom of the reservoir **130**. The fluid **50** in the reservoir **130** will be forced under the second end of the whirlpool shield **175**.

(46) In FIG. **4A** the linear coolant release valve **160A** may include a valve frame **461** and a linear slide door **463** configured to move up and down for opening and closing the linear coolant release valve **160A** and thus changing the size of the exit opening leading to the outlet duct **138** below the whirlpool shield **175**. Actuation of the sliding motion of the linear slide door **463** may be controlled by a linear servo **465**, which may in-turn be controlled by a controller (e.g., **120** and/or **150**, **150-n**). Optionally, a lower edge of the linear slide door **463** may not be straight or at least leave some form of gap, which allows at least a small amount of coolant **50** exit flow C.sub.OUT-2 to leak through the linear coolant release valve **160A** in the fully closed position.

(47) In FIG. **4B** the rotary coolant release valve **160B** may include a valve frame **462** and a rotary pivot door **464** configured to rotate or pivot around a vertical axis for opening and closing the rotary coolant release valve **160B** and thus changing the size of the exit opening leading to the outlet duct **138** below the whirlpool shield **175**. Alternatively, the rotary pivot door **464** could rotate or pivot about a horizontal axis, such as one perpendicular to the exit flow C.sub.OUT-2. Actuation of the rotating or pivotal motion of the rotary pivot door **464** may be controlled by a rotary servo **466**, which may in-turn be controlled by a controller (e.g., **120** and/or **150**, **150-n**). Optionally, the rotary pivot door **464** may not fully close off the opening in the valve frame **462**, leaving some form of gap, which allows at least a small amount of coolant **50** to leak through the rotary coolant release valve **160B** in the fully closed position.

(48) In the embodiments illustrated in FIGS. **4A** and **4B**, the coolant release valves **160A**, **160B** that regulate the flow of dielectric coolant fluid out of the reservoir **130** may be disposed inside a portion of the reservoir **130**. Alternatively, the coolant release valves **160A**, **160B** may be disposed outside the reservoir **130**, such as between the reservoir **130** and shared manifold.

(49) FIG. **5** is a process flow diagram of a method **500** for controlling coolant levels in two or more component cooling tanks sharing a unified coolant distribution system in accordance with various embodiments. With reference to FIGS. **1A-5**, means for performing each of the operations of the method **500** may be a processor of the controller (e.g., **120**, **150**, **150-1**, **150-2**) of the unified coolant distribution system and/or another processor associated with the unified coolant distribution system, and/or the like.

(50) In block **510**, the processor may receive a temperature measurement of a coolant in a first reservoir (e.g., **130-1**) and a second reservoir (e.g., **130-2**) of the two or more component cooling tanks (e.g., **110**). The received temperature measurements from the first reservoir (e.g., **130-1**) and second reservoir (e.g., **130-2**) may not be the same. The received temperature measurements may anticipate fluid level changes, which are a function of temperature changes from a known fluid level.

(51) In some embodiments after receiving the temperature measurement in block **510**, the processor may receive a coolant level measurement of the coolant in the first reservoir and/or the second reservoir in block **512**. Alternatively, the processor may receive the coolant level measurement(s) before receiving the temperature measurement in block **510**. The processor may use the received coolant level measurement(s) to determine the target coolant level update. For example, the

received coolant level measurement(s) may be used to confirm a coolant level determined from the measured coolant temperature.

(52) In block **514**, the processor may signal to a coolant level regulator component (e.g., **145**) of each of the two or more component cooling tanks (e.g., **110**), a target coolant level update. The target coolant level update may be configured to equalize levels of the coolant in the first reservoir and second reservoir to a target coolant level. Also, the target coolant level may be determined based on a temperature average of the received temperature measurements of the coolant in the first reservoir and the second reservoir. In some embodiments, signaling the target coolant level update includes directly activating at least one of a first coolant release valve (e.g., **160, 160-1**) that regulates the release of coolant from the first reservoir or a second coolant release valve (e.g., **160, 160-2**) that regulates the release of coolant from the second reservoir. In some embodiments, signaling the target coolant level update(s) may include using updated coolant temperature measurements (e.g., from the temperature sensor **140, 140-1, 140-2**) and/or coolant level measurements (e.g., from the coolant level detector **170, 170-1, 170-2**) to determine whether coolant should be released from the first and/or second reservoir to meet the coolant level indicated in the target coolant level update.

(53) In some embodiments, the target coolant level update signaled by the processor may be determined by converting the temperature average of the received temperature measurements to a density average of the coolant in the first reservoir and the second reservoir. In some embodiments, signaling the target coolant level update may include signaling a first coolant level controller (e.g., **150, 150-1**) configured to control a first coolant release valve that regulates the release of coolant (e.g., **40, 50**) from the first reservoir, as well as signaling a second coolant level controller (e.g., **150, 150-2**) configured to control a second coolant release valve (e.g., **160, 160-2**) that regulates the release of coolant from the second reservoir. The first coolant level controller may use the received target coolant level update, in conjunction with updated coolant temperature measurements (e.g., from the temperature sensor **140, 140-1**) and/or coolant level measurements (e.g., from the coolant level detector **170, 170-1**) to determine whether coolant should be released from the first reservoir to meet the coolant level indicated in the received target coolant level update. Similarly, the second coolant level controller may use the received target coolant level update, in conjunction with updated coolant temperature measurements (e.g., from the temperature sensor **140, 140-2**) and/or coolant level measurements (e.g., from the coolant level detector **170, 170-2**) to determine whether coolant should be released from the second reservoir to meet the coolant level indicated in the received target coolant level update. In some embodiments, signaling the target coolant level update includes directly activating at least one of a first coolant release valve that regulates the release of coolant from the first reservoir or a second coolant release valve that regulates the release of coolant from the first reservoir.

(54) The operations of the method **500** may be performed continuously or periodically to adjust, maintain, and/or control coolant levels in the two or more component cooling tanks sharing the unified coolant distribution system.

(55) Implementation examples are described in the following paragraphs. While some of the following implementation examples are described in terms of an example computing device memory system, further example implementations may include: the example functions of the computing device memory system discussed in the following paragraphs implemented as methods of the following implementation examples; and the example computing device memory system discussed in the following paragraphs implemented by a computing device memory system including means for performing functions of the computing device memory system of the following implementation examples.

(56) Example 1. A system for controlling coolant levels in a plurality of component cooling tanks sharing a unified coolant distribution system, comprising: first coolant level regulator components of a first component cooling tank comprising: a first temperature sensor configured to measure a

temperature of the coolant in a first reservoir, a first coolant release valve configured to regulate the release of coolant from the first reservoir, and a first coolant level detector configured to detect a level of the coolant in the first reservoir; second coolant level regulator components of a second component cooling tank comprising: a second temperature sensor configured to measure a temperature of the coolant in a second reservoir, a second coolant release valve configured to regulate the release of coolant from the second reservoir, and a second coolant level detector configured to detect a level of the coolant in the second reservoir; and a processor configured to receive the temperature measurements from the first temperature sensor and the second temperature sensor, wherein the processor is configured to signal at least one target coolant level update for adjusting a level of the coolant in the first reservoir by the first coolant release valve, wherein the at least one target coolant level update is determined based on a temperature average of the received temperature measurements from the first temperature sensor and the second temperature sensor, wherein the first coolant release valve is configured to adjust a release of the coolant from the first reservoir in response to comparing the detected level of the coolant in the first reservoir to a target coolant level indicated by the at least one target coolant level update.

(57) Example 2. The system of example 1, wherein the first coolant level detector includes an ultrasonic fluid level sensor.

(58) Example 3. The system of any one or more of examples 1 or 2, wherein the first coolant level regulator components further comprise: a first coolant level controller configured to receive the signaled at least one target coolant level update, wherein the first coolant level controller is configured to activate the first coolant release valve in response to receiving the at least one target coolant level update.

(59) Example 4. The system of example 3, wherein the first coolant level controller is configured to operate independent of the processor configured to receive the temperature measurements from the first temperature sensor and the second temperature sensor.

(60) Example 5. The system of any one or more of examples 1-4, wherein the signaled at least one target coolant level update from the processor is configured to directly control at least the first coolant release valve for regulating the release of coolant associated therewith.

(61) Example 6. The system of any one or more of examples 1-5, wherein the signaled at least one target coolant level update from the processor is determined by converting the temperature average of the received temperature measurements to a density average of the coolant in the first reservoir and the second reservoir.

(62) Example 7. The system of any one or more of examples 1-6, wherein the first coolant release valve includes a linear actuator configured to control a state of a coolant gate regulating the release of coolant.

(63) Example 8. The system of any one or more of examples 1-7, wherein the first coolant release valve includes a rotary actuator configured to control a state of a coolant gate regulating the release of coolant.

(64) Example 9. The system of any one or more of examples 1-8, wherein the processor is further configured to signal the at least one target coolant level update for adjusting a level of the coolant in at least one of the second reservoir by at least the second coolant release valve, wherein the at least one target coolant level update is determined based on the temperature average of the received temperature measurements from the first temperature sensor and the second temperature sensor.

(65) Example 10. The system of example 9, further comprising: third coolant level regulator components of a third component cooling tank comprising: a third temperature sensor configured to measure a temperature of the coolant in a third reservoir, a third coolant release valve configured to regulate the release of coolant from the third reservoir, and a third coolant level detector configured to detect a level of the coolant in the third reservoir; fourth coolant level regulator components of a fourth component cooling tank comprising: a fourth temperature sensor configured to measure a temperature of the coolant in a fourth reservoir, a fourth coolant release

valve configured to regulate the release of coolant from the fourth reservoir, and a fourth coolant level detector configured to detect a level of the coolant in the fourth reservoir, wherein the processor is configured to receive the temperature measurements from the third temperature sensor and the fourth temperature sensor, wherein the at least one target coolant level update signaled by the processor is configured to adjust a level of the coolant in at least one of the third reservoir or the fourth reservoir by the third coolant release valve or the fourth coolant release valve, respectively.

(66) Example 11. The system of any one or more of examples 1-10, wherein the first reservoir and the second reservoir are configured to hold an overflow of the coolant from the first component cooling tank and the second component cooling tank, respectively.

(67) Example 12. The system of any one or more of examples 1-11, wherein the first reservoir and the second reservoir are each the largest coolant reservoirs of the first component cooling tank and the second component cooling tank, respectively.

(68) Example 13. A method for controlling coolant levels in two or more component cooling tanks sharing a unified coolant distribution system, comprising: receiving, at a processor of the unified coolant distribution system, temperature measurements of a coolant in a first reservoir and a second reservoir of the two or more component cooling tanks; receiving, at the processor of the unified coolant distribution system, a coolant level measurement of the coolant in the first reservoir; and signaling, by the processor to a coolant level regulator component of each of the two or more component cooling tanks, a target coolant level update configured to equalize levels of the coolant in the first reservoir and the second reservoir to a target coolant level, wherein the target coolant level is determined based on the received coolant level measurement and a temperature average of the received temperature measurements of the coolant in the first reservoir and the second reservoir.

(69) Example 14. The method of example 13, wherein the target coolant level update signaled by the processor is determined by converting the temperature average of the received temperature measurements to a density average of the coolant in the first reservoir and the second reservoir.

(70) Example 15. The method of any one or more of examples 13 or 14, wherein signaling the target coolant level update includes signaling a first coolant level controller configured to control a first coolant release valve that regulates the release of coolant from the first reservoir; and a second coolant level controller configured to control a second coolant release valve that regulates the release of coolant from the first reservoir.

(71) Example 16. The method of any one or more of examples 13-5, wherein signaling the target coolant level update includes directly activating at least one of: a first coolant release valve that regulates the release of coolant from the first reservoir; or a second coolant release valve that regulates the release of coolant from the first reservoir.

(72) The foregoing descriptions of systems, devices, and methods are provided merely as illustrative examples and are not intended to require or imply that the steps of the various embodiments must be performed in the order presented. As will be appreciated by one of skill in the art the order of steps in the foregoing embodiments may be performed in any order. Words such as “thereafter,” “then,” “next,” etc. are not intended to limit the order of the steps; these words are used to guide the reader through the description of the methods. Further, any reference to claim elements in the singular, for example, using the articles “a,” “an” or “the” is not to be construed as limiting the element to the singular.

(73) The preceding description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the present invention. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other embodiments without departing from the spirit or scope of the invention. Thus, the present invention is not intended to be limited to the embodiments shown herein but is to be accorded the widest scope consistent with the following claims and the principles and novel features disclosed herein.

Claims

1. A system for controlling coolant levels in a plurality of component cooling tanks sharing a unified coolant distribution system, comprising: first coolant level regulator components of a first component cooling tank comprising: a first temperature sensor configured to measure a temperature of the coolant in a first reservoir, a first coolant release valve configured to regulate the release of coolant from the first reservoir, and a first coolant level detector configured to detect a level of the coolant in the first reservoir; second coolant level regulator components of a second component cooling tank comprising: a second temperature sensor configured to measure a temperature of the coolant in a second reservoir, a second coolant release valve configured to regulate the release of coolant from the second reservoir, and a second coolant level detector configured to detect a level of the coolant in the second reservoir; and a processor configured to receive the temperature measurements from the first temperature sensor and the second temperature sensor, wherein the processor is configured to signal at least one target coolant level update for adjusting a level of the coolant in the first reservoir by the first coolant release valve, wherein the at least one target coolant level update is determined based on a temperature average of the received temperature measurements from the first temperature sensor and the second temperature sensor, wherein the first coolant release valve is configured to adjust a release of the coolant from the first reservoir in response to comparing the detected level of the coolant in the first reservoir to a target coolant level indicated by the at least one target coolant level update.
2. The system of claim 1, wherein the first coolant level detector includes an ultrasonic fluid level sensor.
3. The system of claim 1, wherein the first coolant level regulator components further comprise: a first coolant level controller configured to receive the signaled at least one target coolant level update, wherein the first coolant level controller is configured to activate the first coolant release valve in response to receiving the at least one target coolant level update.
4. The system of claim 3, wherein the first coolant level controller is configured to operate independent of the processor configured to receive the temperature measurements from the first temperature sensor and the second temperature sensor.
5. The system of claim 1, wherein the signaled at least one target coolant level update from the processor is configured to directly control at least the first coolant release valve for regulating the release of coolant associated therewith.
6. The system of claim 1, wherein the signaled at least one target coolant level update from the processor is determined by converting the temperature average of the received temperature measurements to a density average of the coolant in the first reservoir and the second reservoir.
7. The system of claim 1, wherein the first coolant release valve includes a linear actuator configured to control a state of a coolant gate regulating the release of coolant.
8. The system of claim 1, wherein the first coolant release valve includes a rotary actuator configured to control a state of a coolant gate regulating the release of coolant.
9. The system of claim 1, wherein the processor is further configured to signal the at least one target coolant level update for adjusting a level of the coolant in at least one of the second reservoir by at least the second coolant release valve, wherein the at least one target coolant level update is determined based on the temperature average of the received temperature measurements from the first temperature sensor and the second temperature sensor.
10. The system of claim 9, further comprising: third coolant level regulator components of a third component cooling tank comprising: a third temperature sensor configured to measure a temperature of the coolant in a third reservoir, a third coolant release valve configured to regulate the release of coolant from the third reservoir, and a third coolant level detector configured to detect a level of the coolant in the third reservoir; fourth coolant level regulator components of a

fourth component cooling tank comprising: a fourth temperature sensor configured to measure a temperature of the coolant in a fourth reservoir, a fourth coolant release valve configured to regulate the release of coolant from the fourth reservoir, and a fourth coolant level detector configured to detect a level of the coolant in the fourth reservoir, wherein the processor is configured to receive the temperature measurements from the third temperature sensor and the fourth temperature sensor, wherein the at least one target coolant level update signaled by the processor is configured to adjust a level of the coolant in at least one of the third reservoir or the fourth reservoir by the third coolant release valve or the fourth coolant release valve, respectively.

11. The system of claim 1, wherein the first reservoir and the second reservoir are configured to hold an overflow of the coolant from the first component cooling tank and the second component cooling tank, respectively.

12. The system of claim 1, wherein the first reservoir and the second reservoir are each the largest coolant reservoirs of the first component cooling tank and the second component cooling tank, respectively.

13. A method for controlling coolant levels in two or more component cooling tanks sharing a unified coolant distribution system, comprising: receiving, at a processor of the unified coolant distribution system, temperature measurements of a coolant in a first reservoir and a second reservoir of the two or more component cooling tanks; receiving, at the processor of the unified coolant distribution system, a coolant level measurement of the coolant in the first reservoir; and signaling, by the processor to a coolant level regulator component of each of the two or more component cooling tanks, a target coolant level update configured to equalize levels of the coolant in the first reservoir and the second reservoir to a target coolant level, wherein the target coolant level is determined based on the received coolant level measurement and a temperature average of the received temperature measurements of the coolant in the first reservoir and the second reservoir.

14. The method of claim 13, wherein the target coolant level update signaled by the processor is determined by converting the temperature average of the received temperature measurements to a density average of the coolant in the first reservoir and the second reservoir.

15. The method of claim 13, wherein signaling the target coolant level update includes signaling: a first coolant level controller configured to control a first coolant release valve that regulates the release of coolant from the first reservoir; and a second coolant level controller configured to control a second coolant release valve that regulates the release of coolant from the first reservoir.

16. The method of claim 13, wherein signaling the target coolant level update includes directly activating at least one of: a first coolant release valve that regulates the release of coolant from the first reservoir; or a second coolant release valve that regulates the release of coolant from the first reservoir.
